CleanGS: Pareto-Guided Post-Hoc Cleanup of 3D Gaussian Splatting Scenes

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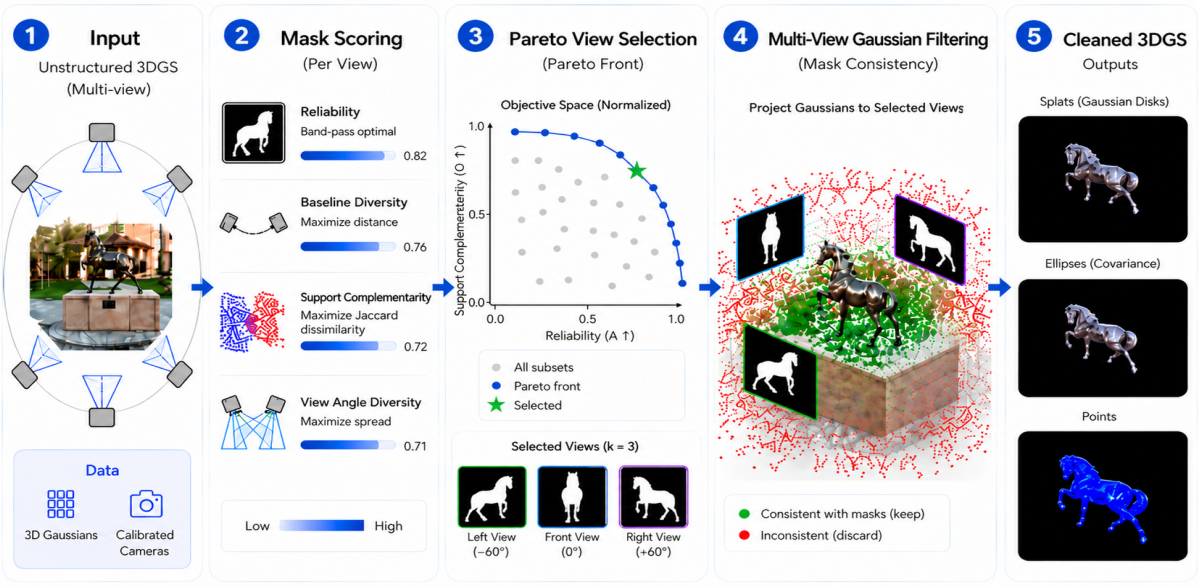


Figure 1: Overview of CleanGS. Candidate masks are scored for reliability, and three-view subsets for camera baseline, viewing-angle diversity, and complementary masked 3D support. We compute the exact Pareto front over the reduced candidate pool, select one balanced subset from it, then apply rasterization-aware two-of-three voting. Retained Gaussians are exported unchanged as a subset of the input scene.

ABSTRACT

A pretrained 3D Gaussian Splatting (3DGS) reconstruction can reproduce its capture views well while retaining off-object structure that becomes visible under reframing. We present CleanGS, a post-hoc cleanup method that removes such structure without retraining the underlying representation. Given calibrated views and binary target masks, CleanGS scores candidate three-view subsets using mask reliability, camera baseline, viewing-direction diversity, and complementary sparse 3D support. Rather than collapsing these objectives a priori, it computes the exact Pareto front over subsets of the reduced candidate pool and selects a balanced non-dominated subset using weighted Chebyshev scalarization. The selected views then drive rasterization-aware two-of-three voting to retain a subset of the original Gaussians. We evaluate the exported asset directly and introduce a fixed-pose widened-field protocol that exposes structure outside the original framing. Across 18 scenes, CleanGS achieves 19.03 dB standard and 17.83 dB widened-field PSNR, compared with 16.44/16.29 dB for FlashSplat and 15.28/15.10 dB for GaussianCut with the same three masks. CleanGS leads five of six photometric metrics and remains within 0.32 dB of the uncleaned reconstruction in standard PSNR. Its filtering-and-export stage takes a median of 20 seconds and 1.9 GB peak memory, compared with 3.1 minutes and 41.6 GB for FlashSplat and 4.2 minutes and 34.9 GB for GaussianCut. Code is available at https://github.com/smlab-niser/clean-gs/.

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CCS CONCEPTS

• **Computing methodologies → Reconstruction**; *Rendering*.

KEYWORDS

3D Gaussian Splatting, scene cleanup, view selection

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1 INTRODUCTION

3D Gaussian Splatting (3DGS) represents a scene as explicit anisotropic Gaussian primitives that support efficient, high-quality novel-view synthesis [12]. Because individual primitives are directly accessible for editing and removal [10], a trained 3DGS scene is also attractive as a reusable 3D asset. However, accurate reconstruction of captured views does not necessarily produce a clean object representation. Photometric optimization can retain primitives outside the object of interest, producing translucent sheets, disconnected fragments, duplicated surfaces, or background structure. We refer to this unwanted retained structure as *off-object clutter*.

Training-time regularization can reduce such artifacts when reconstruction itself can be modified [24, 25, 29]. We instead consider a pretrained scene that must be cleaned post hoc. Binary object masks provide a natural semantic specification of what should remain, but lifting them to Gaussians is ambiguous: a primitive projected inside a target silhouette may belong either to the target or to background along the same viewing ray. Reliable cleanup therefore requires multiview evidence.

The quality of this evidence depends strongly on the chosen views. Some masks provide clear, well-contained observations, while others are truncated, close-up, weakly covering, or geometrically redundant. A small set of reliable and complementary views can therefore provide stronger evidence than treating all observations equally. This motivates our central question: **which views should provide the evidence used to decide which Gaussians belong to the target?**

Our CleanGS couples view selection with Gaussian filtering, as summarized in Figure 1. Given calibrated views and target masks, it scores three-view subsets using mask reliability, camera baseline, viewing-direction diversity, and complementary sparse 3D support. After reducing the candidate pool, we compute the exact Pareto front over these objectives and select a balanced non-dominated subset using a weighted Chebyshev criterion. The selected masks then drive rasterization-aware two-of-three voting, and the surviving Gaussians are exported without modifying their geometry, opacity, or appearance. The three selected views define the filtering budget; the selector may examine a larger pool of candidate masks.

This formulation makes the quality of multiview evidence explicit. Three views provide redundancy for a two-of-three decision, while the selection objectives discourage repeated semantic and geometric evidence. Holding the filtering rule fixed, Pareto-selected views improve both standard and widened-field photometric fidelity over random three-view subsets.

Evaluation presents a second challenge: standard capture views measure target preservation but may hide retained structure outside the original framing. We therefore evaluate the exported Gaussian asset directly and introduce a *fixed-pose widened-field evaluation*. The focal length is reduced while camera pose and image resolution remain fixed, and the same image-plane transformation is applied to the reference image and target mask. This exposes previously out-of-frame structure without requiring ground-truth images from arbitrary novel poses. We report masked PSNR, SSIM, and LPIPS together with silhouette leakage and Gaussian retention.

Across 18 scenes, CleanGS achieves 19.03 dB PSNR at the captured framing and 17.83 dB after widening. At the same three-mask filtering budget, FlashSplat [23] reaches 16.44 and 16.29 dB, while GaussianCut reaches 15.28 and 15.10 dB. An all-mask GaussianCut† reference reaches 17.27 and 16.71 dB. Overall, CleanGS obtains the best result on five of six reported photometric metrics, showing that complementary view selection can preserve target appearance while removing off-object structure without retraining. Its filtering-and-export stage takes 20 seconds (41 seconds end to end, including view selection) and 1.9 GB peak memory, against 3.1 and 4.2 minutes and 41.6 and 34.9 GB for FlashSplat and GaussianCut.

Our contributions are:

(1) a retraining-free cleanup pipeline combining Pareto-guided view selection with rasterization-aware multiview filtering to export a reusable subset of the original Gaussian scene;
(2) a Pareto-guided multi-objective view-selection formulation that exposes the trade-offs among mask reliability, camera baseline, viewing-direction diversity, and complementary sparse 3D support before selecting three views;
(3) a retained-asset evaluation protocol measuring target fidelity at both captured and widened framing together with silhouette leakage and Gaussian retention; and
(4) an 18-scene evaluation showing improved photometric fidelity over retained-asset baselines and random view selection, with analyses of view budget, voting thresholds, and geometric refinement.

2 RELATED WORK

Artifact suppression and pruning. Floating and transient structures can arise when photometric optimization explains the capture images using geometry that is poorly constrained away from the observed views. Recent methods address this problem during reconstruction by introducing visibility, depth, frequency, or robustness priors [24, 25, 29]. These approaches are complementary to our setting, where the Gaussian scene is already trained and cleanup is performed post hoc. Related work on 3DGS compression, including LightGaussian and Compact3DGS, removes primitives according to rate–distortion objectives [8, 14]. Such criteria are effective for reducing representation size, but global reconstruction importance does not by itself distinguish the object of interest from surrounding structure that remains useful for reproducing the captured views. Our setting therefore combines post-hoc pruning with an explicit semantic target.

Mask lifting and object extraction. Recent surveys organize Gaussian scene understanding methods into feature distillation, 2D mask



Figure 2: Ordinary framing hides retained clutter. Top: captured view. Bottom: 6× wider field of view at the same pose. The uncleaned asset reveals substantial off-object structure, while FlashSplat and GaussianCut remove the monument's plinth. CleanGS preserves the plinth while remaining clean after widening. GaussianCut† denotes the all-mask reference.

lifting, and feed-forward prediction [5, 10]. Feature-distillation approaches embed semantic information into Gaussian attributes to support tasks such as open-vocabulary querying [15, 33], while Gaussian Grouping learns per-Gaussian identity features for instance-level interaction [30]. A second family directly lifts multiview 2D masks to the Gaussian representation [3, 6, 34], and CleanGS is most closely related to this line of work.

FlashSplat formulates mask-to-Gaussian assignment as a global linear program [23], while Lifting by Gaussians and LUDVIG study training-free mechanisms for lifting and consolidating 2D evidence in Gaussian scenes [4, 19]. GaussianCut constructs an adjacency graph over Gaussians, converts multiview mask evidence into unary terms, and obtains the object through graph-cut optimization [11]. Our focus is specifically on producing a cleaned object asset whose retained primitives can be exported and rendered directly. For this reason, we evaluate the resulting Gaussian subset rather than a method-specific semantic or label rendering. GaussianCut and FlashSplat provide our primary retained-asset comparisons because both ultimately produce a pruned Gaussian set that can be evaluated under the same rendering protocol as CleanGS. In contrast to applying the lifting procedure to a fixed collection of observations, CleanGS first selects a small set of reliable and complementary views and then uses only their evidence for multiview filtering.

View selection. View selection is well established in multiview geometry, where camera baseline and viewing angle influence triangulation conditioning, and geometrically diverse observations often provide more useful constraints than highly redundant ones [9, 22]. Our setting differs in that the cameras are already calibrated and each candidate view additionally provides a semantic mask and sparse Structure-from-Motion support [21]. We therefore treat view selection as a multi-objective problem that considers both geometric diversity and the quality of the semantic evidence available in each view.

Rather than combining these heterogeneous criteria into a single score at the outset, we evaluate candidate subsets using separate objectives and retain their non-dominated solutions [7, 20]. This formulation makes the trade-offs between semantic reliability, camera separation, viewing direction, and complementary masked 3D support explicit before selecting the subset used for cleanup.

3 METHOD

3.1 Problem Definition

Let $\mathcal{G} = \{g_i\}_{i=1}^{N}$ be a pretrained 3DGS scene, where each Gaussian g_i contains position, rotation, scale, opacity, and spherical-harmonic appearance parameters. The candidate view set $\mathcal{V}$ contains calibrated cameras and binary target masks $\{M_v\}_{v\in\mathcal{V}}$. Camera v has center p_v and unit viewing direction d_v. Our goal is to select a subset $S \subseteq \mathcal{V}$ with $|S| = k = 3$ and use its mask evidence to construct a keep-set $\mathcal{K} \subseteq \{1, \ldots, N\}$. The cleaned asset is obtained by exporting only the Gaussians indexed by $\mathcal{K}$. Figure 1 summarizes the complete pipeline.

We intentionally use a small view budget. Although two calibrated views are sufficient to triangulate visible structure when their viewing rays are not parallel [9, 17], semantic cleanup poses a different problem. A mask provides uncertain evidence about whether a projected Gaussian belongs to the target, and a two-view decision has little redundancy against an erroneous or incomplete mask. We therefore use three views with a two-of-three voting rule. This introduces one redundant observation, allowing a Gaussian to remain supported when one selected mask fails to capture it correctly.

The choice $k = 3$ also provides a favorable balance between filtering quality and exact subset selection. Proposition 3.2 shows that, under conditional independence, two-of-three voting suppresses background support quadratically in the per-view false-support probability. This motivates selecting views whose evidence is as complementary as possible. The supplementary evaluates view budget and vote threshold jointly. The unanimous $k = 2$ configuration is substantially weaker than the main two-of-three setting, while the strongest larger-budget setting, three-of-five voting, matches standard PSNR over all 18 scenes while improving widened PSNR by 0.22 dB and lowering leakage from 0.206 to 0.180.

3.2 Candidate Reliability

We first estimate how suitable each candidate view is for semantic lifting. For view v, let the foreground occupancy be

$$r_v = \frac{|M_v|}{|\Omega_v|}. \tag{1}$$

View quality is not monotonic in occupancy: very small masks provide weak target support, while near-full-frame masks often correspond to close or truncated views. We therefore use the band-pass reliability score

$$a_v = \exp\left[-\frac{1}{2}\left(\frac{r_v - r_0}{\sigma}\right)^2\right], \tag{2}$$

with $r_0 = 0.30$ and $\sigma = 0.15$ for all scenes, favoring intermediate target occupancy. Sensitivity to the occupancy target and to the shape of the reliability score is evaluated in the supplementary.

We further reject poorly contained masks using occupancy, border-contact, and bounding-box tests. When fewer than three candidates survive the subsequent reduction, we fall back to retaining all eligible masks; if selection still fails, we disable the containment tests and cap the pool deterministically at 60 views. The exact thresholds and the capping rule are given in the supplementary.

Each candidate is also associated with sparse 3D support. Let Q_v denote the SfM tracks whose observation in view v lies inside M_v; these provide an inexpensive approximation to the reconstructed support covered by the mask. Before subset enumeration, we remove candidates dominated in reliability, border margin, and truncation, and write $\mathcal{V}_r \subseteq \mathcal{V}$ for the remaining pool. This reduces clearly inferior views while retaining meaningful trade-offs for subsequent subset selection.

3.3 Pareto-Guided Subset Selection

After candidate reduction, we evaluate each feasible subset $S \subseteq \mathcal{V}_r$ using both semantic quality and geometric complementarity. For an unordered pair set $P(S) = \{(u, v) : u, v \in S,\ u < v\}$, we define

$$A(S) = \min_{v \in S} a_v, \tag{3}$$

$$B(S) = \frac{1}{|P(S)|} \sum_{(u,v)\in P(S)} \|p_u - p_v\|_2, \tag{4}$$

$$O(S) = \frac{1}{|P(S)|} \sum_{(u,v)\in P(S)} \left(1 - \frac{|Q_u \cap Q_v|}{|Q_u \cup Q_v|}\right), \tag{5}$$

$$T(S) = \frac{1}{|P(S)|} \sum_{(u,v)\in P(S)} \arccos(\langle d_u, d_v \rangle). \tag{6}$$

The four objectives capture distinct properties of a useful subset. $A(S)$ measures the reliability of its weakest view, since one poor mask can systematically affect the subsequent vote. $B(S)$ favors separation between camera centers. $O(S)$ is a Jaccard *dissimilarity* over masked SfM tracks and is therefore maximized. It discourages subsets whose views encode nearly identical sparse support. It carries a low weight, is balanced against A and the geometric terms, and the final two-of-three vote still requires cross-view agreement before a Gaussian is retained. $T(S)$ favors diversity in viewing direction, and we retain it alongside $B(S)$ because a large camera displacement does not necessarily imply a large change in viewing direction.

These criteria have different scales and encode different notions of desirable evidence, so we do not combine them into a single score before enumerating their trade-offs. We instead evaluate all feasible k-subsets $S \subseteq \mathcal{V}_r$ and compute their exact Pareto front under $f(S) = (A, B, O, T)$. A subset is dominated when another subset is at least as good in every objective and strictly better in at least one. Dominated subsets cannot achieve a strictly larger value under any utility that is non-decreasing in all four objectives.

We min–max normalize the objectives and select a single point from the exact front using weighted Chebyshev distance to the fixed ideal $z = (1, 1, 1, 1)$:

$$S^\star = \arg\min_{S\in\mathcal{F}} \max_{\ell} w_\ell \big(z_\ell - \widetilde{f}_\ell(S)\big), \tag{7}$$

where $\mathcal{F}$ denotes the exact Pareto front. Chebyshev scalarization penalizes solutions whose weakest normalized objective remains far from the ideal [28]. Since the scalarization is monotone in the objectives, Lemma 3.1 guarantees that its optimal value is attained on $\mathcal{F}$. The front also keeps the non-dominated trade-offs explicit, which allows the same objective space to be examined under different weightings, as we do in the supplementary.

We use weights $(0.45, 0.30, 0.15, 0.10)$ for (A, B, O, T) in every scene. The supplementary provides further sensitivity experiments and a discussion on weight choices. The weighting emphasizes A and B, which account for most selection changes at this operating point: reliability controls the semantic evidence entering the vote, and baseline provides a strong geometric cue against redundant observations. The lower-weight O and T terms make smaller but still measurable contributions: zeroing either reduces standard PSNR by 0.05–0.08 dB and widened-field PSNR by 0.16–0.22 dB, while also increasing leakage. Their effect is more pronounced under widened framing, where redundant view selection is more likely to leave residual structure exposed. Across the tested settings, standard PSNR varies by at most 0.14 dB, while placing greater emphasis on the diversity terms can further improve the widened-field operating point.

Figure 3 shows the selected subsets for three representative scenes.

Lemma 3.1 (Dominance is preference-free). *Let $U(S) = \Phi(f(S))$ for any Φ that is non-decreasing in each of its four arguments. If S_2 dominates S_1, then $U(S_2) \geq U(S_1)$. Consequently $\max_S U(S)$ is attained on $\mathcal{F}$.*

The proof and a discussion of each objective are in the supplementary. The Chebyshev weights therefore only choose among mutually non-dominated alternatives.

3.4 Multiview Filtering and Export

The selected subset $S^\star$ provides the mask evidence used to filter the Gaussian scene. We rasterize $\mathcal{G}$ from each selected camera and use renderer visibility to exclude Gaussians that cannot contribute to the image. With projection π_v and visibility vis_v, we define

$$X_v(g_i) = \mathbf{1}[\mathrm{vis}_v(g_i)]\,\mathbf{1}[\pi_v(\mu_i) \in M_v], \quad \nu_i = \sum_{v\in S^\star} X_v(g_i). \tag{8}$$

For $k = 3$, we retain g_i when $\nu_i \geq 2$. This two-of-three rule requires multiview agreement while tolerating one incomplete or inconsistent mask. The benefit of voting depends on the selected views providing complementary evidence. The following proposition characterizes the idealized case of conditionally independent support.

Proposition 3.2 (Independent support suppresses clutter). *Let $p_f = \Pr[X_v(g) = 1 \mid g \text{ foreground}]$ and $p_b = \Pr[X_v(g) =$*

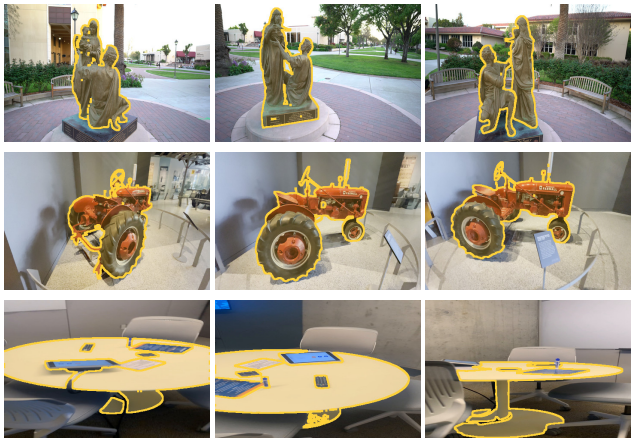


Figure 3: Selected three-view subsets used for Gaussian filtering $S^\star$ for *Family*, *tractor*, and *table*, with masks outlined. The selected views balance mask reliability with geometric and support diversity.

Algorithm 1 CLEANGS

Require: Gaussians $\mathcal{G}$, calibrated views $\mathcal{V}$ with masks $\{M_v\}$, sparse tracks, subset size $k = 3$, threshold $\tau = 2$
Ensure: Retained Gaussians $\mathcal{G}[\mathcal{K}]$
score each v: occupancy r_v, band-pass a_v, masked track set Q_v, and reject truncated candidates
reduce the pool by mask-level dominance, then cap it, giving $\mathcal{V}_r$
for all $S \subseteq \mathcal{V}_r$ with $|S| = k$ **do**
    evaluate $f(S) = (A, B, O, T)$
end for
$\mathcal{F} \leftarrow$ exact Pareto front of $\{f(S)\}$ ▷ Lemma 3.1
$S^\star \leftarrow \arg\min_{S \in \mathcal{F}} \max_\ell w_\ell(z_\ell - \widetilde{f}_\ell(S))$
for all $v \in S^\star$ **do**
    rasterize $\mathcal{G}$ and record visibility and projected centers
end for
$\nu_i \leftarrow \sum_{v \in S^\star} X_v(g_i)$
$\mathcal{K} \leftarrow \{i : \nu_i \geq \tau\}$ ▷ Prop. 3.2
return $\mathcal{G}[\mathcal{K}]$, attributes unchanged

$1 \mid g \text{ background}]$ *with* $p_f > p_b$, *and suppose the* k *indicators* $\{X_v(g)\}_{v \in S^\star}$ *are conditionally i.i.d. given the class of* g. *Under the keep rule* $\nu(g) \geq \tau$,

$$\Pr[\nu(g) \geq \tau \mid g] = \sum_{j=\tau}^{k} \binom{k}{j} p^j (1-p)^{k-j}, \tag{9}$$

with $p = p_f$ *for foreground and* $p = p_b$ *for background. For our* $k = 3$ *and* $\tau = 2$,

$$\Pr[\nu(g) \geq 2 \mid g \text{ background}] = 3p_b^2(1-p_b) + p_b^3 \leq 3p_b^2. \tag{10}$$

The proof is in the supplementary. Under conditional independence, background survival is quadratic in p_b for small p_b, motivating B, T, and O to discourage repeated support, while A favors reliable foreground evidence. Real views are correlated: on our scenes, two-of-three agreement is about twice as frequent as independence predicts (median; supplementary), so the proposition is an idealization. Increasing the number of votes alone therefore does not necessarily provide stronger evidence; the selector instead seeks a small set of reliable and complementary observations.

Voting is most effective for clutter that is separated from the target and supported by few masks. Target-adjacent structure is harder, because it can stay aligned with the target across several views; Section 5.5 evaluates geometric refinement for this regime.

Algorithm 1 summarizes the complete procedure. Cleanup is purely subtractive: the output contains only Gaussians from the original scene, with their geometry, opacity, and appearance unchanged, and requires no gradient-based optimization.

For n candidates after reduction, subset scoring considers $\binom{n}{3}$ triples, while all-pairs construction of the exact Pareto front costs $O(\binom{n}{3}^2)$ and dominates selection. Candidate reduction is therefore important for keeping exact search practical. By Lemma 3.1, minimizing the Chebyshev criterion directly over all triples returns the same $S^\star$ in $O(\binom{n}{3})$ time; we build $\mathcal{F}$ only to expose the trade-offs. Once $S^\star$ is selected, filtering requires only three rasterization passes and is linear in scene size. The supplementary gives the full per-stage complexity analysis and empirical subset counts.

4 RETAINED-ASSET EVALUATION

4.1 Why Widen the Field of View?

A cleaned Gaussian scene should preserve the appearance of the target while removing structure that should not remain in the exported asset. Standard capture views measure the first property well, but provide only limited evidence about the second. Gaussians outside the original image boundaries do not contribute to image-space metrics even when they remain part of the scene. Evaluating arbitrary novel viewpoints would expose more of this structure, but existing datasets generally do not provide corresponding ground-truth RGB observations.

We therefore introduce a controlled fixed-pose reframing test. For each evaluation camera, we keep the camera pose and image resolution unchanged and divide the focal length by six. We apply the corresponding image-plane transformation to the reference RGB image and target mask, and evaluate every method using the same transformed camera. Photometric metrics use only target pixels that have captured reference RGB, while leakage also counts newly exposed off-target pixels. The resulting view exposes a larger portion of the retained scene while preserving the original camera center and orientation.

This transformation is used only for evaluation. It does not modify the Gaussian representation or introduce new geometry, but changes which parts of the fixed scene fall inside the image. We refer to this setting as the *widened-field* or *zoom-6* evaluation. The results are stable to the magnitude of the focal-length change: at 2× and 4× widening, CLEANGS remains the strongest cleanup method in PSNR and the ordering among the three cleanup approaches is unchanged. The supplementary reports all four focal-length settings.

4.2 Metrics

Let I_v denote the reference image, $\widehat{I}_v$ the rendered image, M_v the target mask, and A_v the rendered opacity for camera v. We first define the masked squared error and the number of evaluated target pixels as

$$E_v = \sum_p M_v(p) \|I_v(p) - \widehat{I}_v(p)\|_2^2, \qquad Z_v = \sum_p M_v(p). \tag{11}$$

We report masked

$$\mathrm{MSE}_M(v) = \frac{E_v}{3Z_v}, \qquad \mathrm{PSNR}_M(v) = 10 \log_{10}(1/\mathrm{MSE}_M(v)), \tag{12}$$

together with masked SSIM [26] and LPIPS [32]. The factor of three averages over the colour channels. These metrics are computed over the target region only, so they are not directly comparable to full-image PSNR reported on object-centric benchmarks; Z_v falls from a mean of 5.8×10^5 pixels at the captured framing to 2.3×10^4 at 6×, a factor of 24.6 rather than the 36 implied by area alone, because some targets are already clipped by the captured frame.

Photometric fidelity alone does not measure how much off-target structure remains. We therefore additionally measure silhouette leakage,

$$\mathcal{L}_{\mathrm{leak}}(v) = \frac{\sum_p (1 - M_v(p))\, A_v(p)}{\sum_p (1 - M_v(p))}, \tag{13}$$

the mean rendered opacity outside the target region. The supplementary also reports a thresholded variant that counts only pixels with $A_v(p) > 0.1$. We also report the fraction of the original Gaussians retained by each method as a measure of asset compactness.

Leakage and retention are both compactness measures and therefore reward more aggressive pruning: a method that deleted everything outside the target could achieve zero leakage even if it damaged the target, while the uncleaned scene scores 0.983 because it retains the complete reconstruction. We therefore report compactness together with photometric fidelity so that lower leakage is interpreted alongside preservation of the target appearance. Unless stated otherwise, aggregate leakage is reported at the captured (1×) framing using the full evaluation masks.

4.3 Data, Protocol, and Comparators

All scenes use pretrained `gsplat` checkpoints [31]. We evaluate across a diverse scene set: 8 DL3DV [16], 7 NeRFBusters [27], and 3 Tanks and Temples [13] scenes, for 18 in total. These standard benchmarks span indoor and outdoor settings, close-up and wider compositions, and casual and more controlled captures. We report per-scene averages using the same 6× fixed-pose widening throughout. Target masks are obtained once from SAM 3 [2] using manual prompting, checked only for target coverage, and reused unchanged for every method. CleanGS selects only from training cameras, keeping evaluation masks unavailable to the selector. Matched GaussianCut uses the same three selected training-view masks and therefore follows the same separation between filtering and scored views.

We compare against the uncleaned 3DGS scene, FlashSplat, and GaussianCut by rendering each exported Gaussian asset directly. FlashSplat uses removal slackness 0 and the same three masks as CleanGS. GaussianCut is evaluated both with these masks and with all available capture masks; the latter, denoted GaussianCut†, serves as a denser-supervision reference. Further baseline details are given in the supplementary. Quantitative comparison is restricted to methods that export a directly renderable Gaussian subset. Semantic-label methods such as Gaussian Grouping and Gaga [18, 30] are discussed further in the supplementary.

5 RESULTS

5.1 Appearance–Cleanup Trade-off

Table 1 summarizes the main retained-asset comparison. CleanGS achieves the best result on five of the six photometric metrics while retaining 22.8% of the original Gaussians. With the same three masks, CleanGS improves over FlashSplat and GaussianCut by 2.59 and 3.75 dB in standard PSNR, and by 1.54 and 2.73 dB under widened framing. Compared with all-mask GaussianCut†, the corresponding margins are 1.76 dB and 1.12 dB.

The only photometric metric led by the uncleaned scene is standard-view PSNR, where the difference is 0.32 dB. CleanGS nevertheless improves over the uncleaned scene in standard SSIM and LPIPS and in all three widened-field metrics. The scene-level results show a similar pattern. CleanGS achieves higher PSNR at both framings on 16 of 18 scenes compared with FlashSplat and on 15 of 18 compared with all-mask GaussianCut†, and on all 18 scenes compared with GaussianCut under matched three-mask supervision. Median PSNR gives the same ranking (supplementary). Standard-view PSNR varies across scenes with a standard deviation of approximately 2.8 dB, and the supplementary reports the complete per-scene results.

The cleanup statistics capture the complementary effect of more aggressive pruning. GaussianCut attains the lowest leakage, with 0.089 under matched supervision and 0.055 when given all capture masks, compared with 0.141 for FlashSplat and 0.206 for CleanGS. FlashSplat is the most compact at 14.9% retention. These gains in silhouette compactness are accompanied by lower target photometric fidelity, particularly under matched supervision. The all-mask GaussianCut result additionally uses masks at evaluation cameras, so we report it as an all-mask reference. Overall, CleanGS occupies a favorable operating point that preserves target appearance while removing more than three quarters of the original Gaussian scene.

5.2 What Widened Framing Reveals

The widened-field evaluation highlights differences that are less apparent at the captured framing. Widening also resamples the reference target, so we compare methods within each framing rather than interpret absolute changes between 1× and 6×. The mechanism behind the widened ranking is nevertheless clear. Reducing the focal length enlarges the view frustum, so Gaussians that were previously culled now enter it. Those lying between the camera and the target project with large screen-space footprints and composite over target pixels, which degrades fidelity inside the mask even though the offending structure is outside it. The uncleaned scene admits the most such structure and moves from first at the captured framing to last at 6×, while cleaned assets have little left to admit. Figure 2 illustrates the same behavior qualitatively.

Standard and widened PSNR measure target fidelity under different framings and are therefore complementary, while widened leakage separately measures off-target structure exposed by the larger

Table 1: Retained-asset comparison on 18 scenes. Standard: captured framing; Widened: 6× fixed-pose widened field of view. All methods use the same scored frames and evaluation masks; FlashSplat, GaussianCut, and CleanGS use the same three selected masks, and GaussianCut† uses all capture masks. Best photometric result is in bold; best cleanup statistic is underlined.

	Masks used	Standard view			Widened field of view			Leakage (1×) ↓	Retained ↓
Method	for cleanup	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓		
Uncleaned 3DGS	0	**19.35**	0.620	0.194	12.84	0.300	0.327	0.983	100.0%
FlashSplat [23]	selected 3	16.44	0.557	0.257	16.29	0.411	0.205	0.141	<u>14.9%</u>
GaussianCut [11]	selected 3	15.28	0.533	0.279	15.10	0.381	0.227	0.089	17.2%
GaussianCut† [11]	all capture	17.27	0.572	0.234	16.71	0.421	0.194	<u>0.055</u>	20.2%
CleanGS (ours)	selected 3	19.03	**0.643**	**0.188**	**17.83**	**0.501**	**0.167**	0.206	22.8%

field of view. Within the captured framing, CleanGS stays within 0.32 dB of the uncleaned reconstruction, whereas GaussianCut† gives up 2.08 dB. Within the widened framing, CleanGS exceeds the uncleaned scene by 4.99 dB and GaussianCut† by 3.87 dB, and CleanGS achieves the strongest PSNR among the cleanup methods. CleanGS therefore preserves much of the target fidelity of the original reconstruction while producing an asset that is substantially more stable under reframing.

5.3 Selector Coverage

We next isolate the contribution of view selection. Train-only candidate pools are available for all 18 scenes, and exact selection completes for every scene at the $k = 3$ operating point used in Table 1.

We hold the voting procedure and evaluator fixed and replace the selected subset with three uniformly sampled training views, using five random seeds per scene and averaging within scene. The paired photometric comparison covers 17 scenes because, on *super markets*, three random subsets retain no Gaussians and therefore have undefined photometrics. Pareto-selected views improve all six photometric metrics. Standard PSNR increases by 1.56 dB, with selection favored on 15 of 17 scenes, while widened PSNR improves by 1.00 dB, favored on 14 of 17. Two-sided Wilcoxon signed-rank tests give p=0.0005 and p=0.015, respectively. The supplementary reports the corresponding SSIM, LPIPS, leakage, and retention results.

Random subsets retain fewer Gaussians and achieve lower leakage on average, but also preserve less target appearance and can fail on difficult scenes. On *super markets*, three of five random subsets remove every Gaussian, whereas CleanGS produces a non-empty output on all 18 scenes.

5.4 Cleanup Cost

Table 2 compares filtering-and-export cost at the same three-mask filtering budget. Filtering and export take a median of 20 seconds for CleanGS, compared with 3.1 minutes for FlashSplat and 4.2 minutes for GaussianCut, approximately 9× and 13× lower runtime. Peak memory is 1.9 GB for CleanGS, compared with 41.6 GB and 34.9 GB, an approximately 18–22× reduction. View selection is a separate CPU stage and is excluded from these numbers; including it, the end-to-end median is 41 seconds (supplementary).

Table 2: Filtering and export cost per scene over 18 scenes at the matched three-mask budget. Times and peak memory exclude view selection.

Method	Time ↓	Range	Peak memory ↓
FlashSplat [23]	3.1 min	0.7–6.2 min	41.6 GB
GaussianCut [11]	4.2 min	1.0–14.4 min	34.9 GB
CleanGS (ours)	**20 s**	12–58 s	**1.9 GB**

The efficiency follows directly from the structure of the pipeline. After view selection, CleanGS requires three rasterization passes and a vote over the resulting support indicators, with no per-Gaussian semantic feature field, no Gaussian adjacency graph, and no global optimization over the full scene. GaussianCut instead builds a graph whose size grows with the Gaussian count and solves a max-flow problem. All runs use an A100-80GB GPU.

5.5 Does Geometric Refinement Help?

The multiview vote removes structure that is inconsistent with the selected masks, but nearby surfaces that remain mask-consistent can survive. We therefore test a simple geometric refinement by applying HDBSCAN [1] to the retained Gaussians and keeping only the largest density component.

Under the eroded evaluation masks used for this paired ablation, clustering reduces leakage from 0.221 to 0.115, showing that spatial connectivity provides useful cleanup signal, but standard and widened PSNR drop by 3.84 dB and 3.28 dB, respectively. Both endpoints use the same eroded masks, so the comparison isolates the clustering stage; its absolute values are therefore not directly comparable with Table 1. Complete aggregate and per-scene results are given in the supplementary.

The fidelity loss is concentrated in scenes where the target spans multiple density components. Six scenes lose more than 3 dB in standard PSNR, while the remaining twelve change only slightly. On *tractor*, for example, the largest of 92 clusters contains only 9% of the retained Gaussians, so keeping a single component removes substantial target structure, indicating that a valid target can occupy several density modes. We therefore use the vote-only output as the CleanGS endpoint.

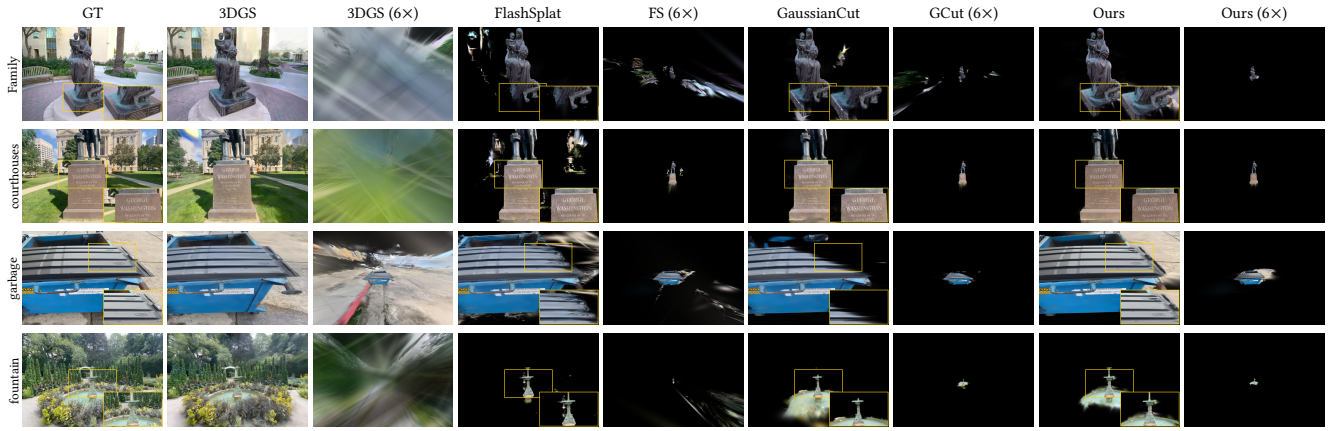


Figure 4: Retained target appearance. Captured and 6× widened renderings use the same camera pose, so widening reveals content already retained in the exported asset. FlashSplat introduces visible target artifacts, while GaussianCut often produces tighter silhouettes at the cost of darker target regions. CleanGS better preserves details such as the plinth, inscription, lettering, and fountain basin.

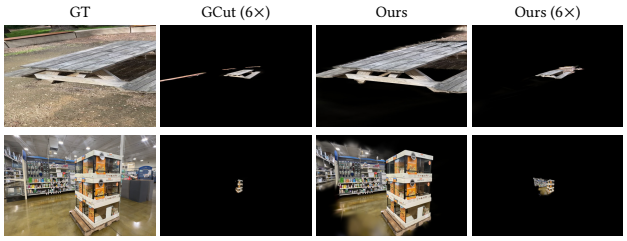


Figure 5: Residual target-adjacent structure. Nearby surfaces in *super markets* and *picnic* remain supported across selected views. Their widened leakage is 0.221 and 0.099, against a benchmark mean of 0.066. GaussianCut† removes these regions with all-mask supervision; on *picnic*, it is 1.49 dB better in widened PSNR.

6 LIMITATIONS

CleanGS preserves target appearance well, but nearby surfaces can survive when they remain consistently aligned with the target across selected views (supplementary). Figure 5 shows two representative failures. In *super markets*, a shelving surface behind the target remains supported across views; in *picnic*, an attached surface survives for the same reason. These cases illustrate the main limitation of semantic voting when target and background remain consistently aligned across viewpoints.

Renderer visibility indicates whether a Gaussian can contribute to an image, but not whether it is geometrically unoccluded by the target. Background along the same viewing ray can therefore receive repeated support. Neither a largest-component prior (Section 5.5) nor simple mask erosion reliably resolves this case: 4-pixel erosion reduces leakage from 0.206 to 0.196 but costs 0.66 dB in standard PSNR. Mask membership is also tested only at the projected Gaussian center, making thin boundaries sensitive to mask error (a footprint-aware variant is evaluated in the supplementary), while transparent and reflective targets may genuinely depend on nearby radiance.

The selector additionally depends on diversity and reconstructed support in the available views. Sparse tracks are less informative for poorly reconstructed targets, limited camera variation weakens the geometric objectives, and selection cannot recover evidence absent from the candidate masks. Finally, fixed-pose widening tests controlled reframing rather than arbitrary camera motion and cannot expose permanently occluded clutter.

7 CONCLUSION

We presented CleanGS, a retraining-free cleanup pipeline for pretrained 3DGS scenes. It selects reliable and complementary views using semantic and geometric objectives, chooses a balanced three-view subset from the exact Pareto front over the reduced candidate pool, and applies rasterization-aware two-of-three filtering while leaving the parameters of retained Gaussians unchanged. Across 18 scenes, CleanGS achieves the best result on five of six reported photometric metrics, with its advantage particularly clear under widened framing, and its filtering-and-export stage runs 9–13× faster with 18–22× lower peak memory than the retained-asset baselines. The random-view comparison further shows that selecting the evidence used for filtering is essential. Future work can combine multiview semantic evidence with stronger occlusion and geometry-aware reasoning for target-adjacent structure and explore sequential selection under an explicit supervision budget.

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CODE AND SUPPLEMENTARY MATERIAL

Code and the supplementary material are available at https://github.com/smlab-niser/clean-gs/tree/main.